ROBOTICS LAB ADMISSION PROJECT

Automatic Classroom People Counter

Real-time occupancy tracking using IR sensors and I2C LCD

PROJECT MEMBERS

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The Problem & Solution

Manual People Counting

- ✕ Extremely slow process.
- ✕ Disrupts lecture flow.
- ✕ Prone to human error.

Automated System

- ✓ 100% automated counting.
- ✓ Zero manual intervention.
- ✓ Instant real-time accuracy.

Project Objective

Automate classroom occupancy tracking via real-time IR detection.

Bi-directional counting

Instant LCD visual output

5V low-power architecture

System Architecture



Input

2x IR Sensors



Processing

Arduino Uno



Output

16x2 I2C LCD

Core Components



Arduino Uno

Main processing and logic control unit.



IR Sensors (x2)

Directional entry and exit motion detection.



16x2 I2C LCD

Real-time visual display (address 0x27).

Circuit Pin Mapping

Peripheral	Module Pin	Arduino Uno Pin
IR Sensor 1 (Entry)	OUT Data	**Digital Pin 2**
IR Sensor 2 (Exit)	OUT Data	**Digital Pin 3**
I2C LCD Module	SDA (Data)	**Analog Pin 4 (A4)**
I2C LCD Module	SCL (Clock)	**Analog Pin 5 (A5)**
All Modules	VCC / GND	**5V / GND**

Detection Logic

Entry Sequence

Sensor 1 → Sensor 2

Count +1

Exit Sequence

Sensor 2 → Sensor 1

Count -1

Software Pipeline

1. Scan

Poll digital pins 2 and 3 continuously.

2. Sequence

Determine which IR sensor triggered first.

3. Update

Adjust interior integer variable (+1 or -1).

4. Render

Push updated string via I2C protocol.

Real-World Applications

Smart Classrooms

Automated student attendance tracking.

Libraries & Malls

Live capacity and crowd management.

Smart Buildings

Occupancy-based HVAC automation.

Public Transport

Real-time passenger load tracking.

Key Advantages



Low Cost

Affordable, off-the-shelf electronic components.



Real-Time

Instant occupancy data processing.



Non-Intrusive

No ID badges or manual scanning required.

Future Enhancements

IoT Integration

Cloud data logging via ESP8266 Wi-Fi module.

Vision AI

Camera-based crowd analysis for high-density entryways.

Hardware Automation

Syncing active occupancy count with room lighting relays.

Thank You

We are open to your questions.